AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS (THEORY) 2025

Business Statistics Paper II

Time: 1 hour 5 minutes Marks: 20

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are SIX questions. Answer ALL questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. A formulae list is provided on page 2. You may refer to it during the paper, if you wish.

6. You may use a scientific calculator if you wish.

List of Formulae

Note:

- The symbols used in the formulae have their usual meanings.

Measures of Central Tendency and Quartiles

$$\overline{X} = \frac{\sum wx}{\sum w} \qquad \overline{X} = \frac{\sum x}{n} \qquad \overline{X} = \frac{\sum fx}{\sum f} \qquad \mathrm{Q}_i = l + \frac{h}{f}\left(\frac{n \times i}{4} - c\right)$$

$$\text{Median} = l + \frac{1}{f}\left(\frac{n}{2} - c\right) \times h \qquad \text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

Measures of Dispersion

$$\text{variance} = \frac{\sum \left(x - \overline{X}\right)^2}{n} \qquad \text{SD} = \sqrt{\frac{\sum \left(x - \overline{X}\right)^2}{n}}$$

$$\text{variance} = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2 \qquad \text{SD} = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

$$\text{variance} = \frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2 \qquad \text{SD} = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2}$$

$$\text{variance} = \frac{\sum f\left(x - \bar{x}\right)^2}{\sum f} \qquad \text{SD} = \sqrt{\frac{\sum f\left(x - \bar{x}\right)^2}{\sum f}}$$

$$\mathrm{R} = x_{\max} - x_{\min} \qquad \text{Coefficient of Variance} = \frac{SD}{\overline{X}} \times 100$$

Index Number

$$P_{0n} = \frac{\sum p_n q_0}{\sum p_0 q_0} \times 100 \qquad P_{0n} = \frac{\sum p_n q_n}{\sum p_0 q_n} \times 100 \qquad P_{0n} = \sqrt{\frac{\sum p_n q_0}{\sum p_0 q_0} \times \frac{\sum p_n q_n}{\sum p_0 q_n}} \times 100$$

Counting Techniques and Probability

$${}^{n}P_r = \frac{n!}{(n-r)!} \qquad {}^{n}C_r = \frac{n!}{(n-r)!r!}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \qquad P(A \cap B) = P(A) \times P(B \mid A)$$

$$P(A \cap B) = P(A) \times P(B) \qquad P(A \cap B) = P(B) \times P(A \mid B)$$

Scatter Diagram and Correlation

$$r = \frac{n \sum XY - \left(\sum X\right)\left(\sum Y\right)}{\sqrt{\left[n \sum X^2 - \left(\sum X\right)^2\right]\left[n \sum Y^2 - \left(\sum Y\right)^2\right]}}$$

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Q.1. (Total 3 Marks)

Complete the given frequency distribution table.

Weight (in pounds)	Class Boundaries	Frequency
	99.5 - 129.5	5
130 - 159	129.5 - 159.5	10
160 - 189		
190 - 219	189.5 - 219.5	7
Total	**-**	**32**

(ATTEMPT EITHER PART a OR PART b OF Q.2.)

Q.2. (Total 3 Marks)

a. In a sports academy, players of different age groups are registered. These age groups are recorded in the given table.

Complete the frequency table to find the average age of the players.

Age Groups	Number of Players (f)		
14 - 17	10		
18 - 21	25		
22 - 25	30		
26 - 29	15		
30 - 33	20		
34 - 37	14		

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(**ATTEMPT EITHER PART a OR PART b OF Q.2.)**

b. Consider the given frequency distribution.

Class intervals	Class boundaries	Frequency	Cumulative Frequency
0 - 9	0.5 - 9.5	5	5
10 - 19	9.5 - 19.5	7	12
20 - 29	19.5 - 29.5	12	24
30 - 39	29.5 - 39.5	10	34
40 - 49	39.5 - 49.5	8	42
50 - 59	49.5 - 59.5	13	55
60 - 69	59.5 - 69.5	10	65

Using the given frequency distribution, calculate the

i. 1st quartile. (2 Marks)

ii. 3rd quartile. (1 Mark)

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Q.3. (Total 3 Marks)

The recorded temperature of a town for 10 days is shown in the given table. The second row shows the difference between daily temperature and mean temperature of the 10 days.

Daily Temperature (X_i)	12	12	14	15	16	17	20	22	22	20
$X_i - \overline{X}$	-5	-5	-3	-2	-1	0	3	5	5	3
$\left(X_i - \overline{X}\right)^2$										

Complete the given table to find the standard deviation and variance in the temperatures.

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Q.4. (Total 4 Marks)

The given table shows the production of cars in a factory is given for the years 1990 to 1995.

Year	**Production of Cars**	**Link Relative**	**Chain Index**
1990	800		
1991	900		
1992	1,100		
1993	1,100		
1994	1,300		
1995	1,500		

Calculate the index number by chain base method using 1990 as a base year.

(ATTEMPT EITHER PART a OR PART b OF Q.5.)

Q.5. (Total 4 Marks)

a. Akram has a box containing the letters *B, R, A, I, N* and *S*.

i. How many arrangements can he form using any four alphabets at a time? (1 Mark)

ii. How many arrangements can he form using all the given alphabets, but *A* and *I* should appear together? (1 Mark)

iii. In how many ways can select the names of the vertices of a triangle? (1 Mark)

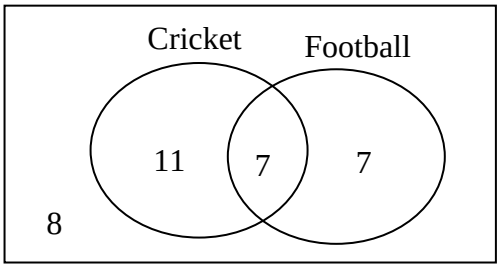
iv. In how many ways can select the names of the vertices of a triangle, if a particular alphabet must be a name for one of the vertices? (1 Mark)

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(ATTEMPT EITHER PART a OR PART b OF Q.5.)

b. Faiz asked 33 students whether they liked watching cricket or football matches on TV.

The result is shown in the given Venn diagram.



If one student is chosen at random, find the probability that s/ he likes watching

i. only cricket. (1 Mark)

ii. cricket or football. (1 Mark)

iii. cricket and football. (1 Mark)

iv. either both or neither. (1 Mark)

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Q.6. (Total 3 Marks)

The data shown in the given table represents the advertisement cost (in thousand rupees) and sales (in hundred thousand rupees) in a company.

Using the given information, calculate the coefficient of correlation.

Advertisement Cost (in Thousand Rupees, *x*)	**Sales (in Hundred Thousands Rupees, *y*)**	x^2	y^2	xy
40	38	1600	1444	1520
39	43	1521	1849	1677
45	38	2025	1444	1710
49	45	2401	2025	2205
48	49	2304	2401	2352
50	47	2500	2209	2350
47	48	2209	2304	2256
44	44	1936	1936	1936
46	45	2116	2025	2070
48	48	2304	2304	2304

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END OF PAPER

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